

UDESH MUKUNDAN

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PROFILE

MSc Biotechnology graduate with strong exposure in clinical instrumentation testing, QC protocols, and bioinformatics. Skilled in wet lab techniques (electrophoresis, chromatography), clinical chemistry testing (AST, ASO, glucose), and bioinformatics automation. Experienced with scripting (Python, Bash) and cloud deployment for research tools.

SKILLS

Wet Lab: Gel Electrophoresis, Spectrophotometer, LC-MS, GC-MS, DNA/RNA Isolation, PCR, Macrophage Culturing, Blood Profiling, Microscopy, Biochemical Analysis, Clinical Instruments: BTS 350 (photometric accuracy, linearity, glucose, AST, ASO tests), Bioinformatics: BLAST, AutoDock, SwissDock, UniProt, NCBI, Sequence Analysis, GATK, Tech Tools: Python, Bash, Django, AWS, Terraform, Docker, Kubernetes, PostgreSQL, MS Excel

EXPERIENCE

QC Intern – Biosystems Pvt. Ltd., Chennai | Feb 2022 – Jun 2022

- Analyzed assembled BTS 350 instruments for absorbance, precision, and clinical chemistry (Glucose, AST, ASO)
- Conducted photometric accuracy and linearity control tests to verify instrument performance
- Documented instrument validation reports under GLP and ISO compliance

Freelance Web Developer, DevOps & Bioinformatics Developer | Remote | 2023 – Present

- Built Django-based lab, clinic, and booking systems with PDF/Excel reports and AWS deployment
- Developed NGS pipelines and neuroreceptor sequence workflows using Bash, Python, and cloud automation
- Automated infrastructure with Terraform and Kubernetes, supporting biotech and research use cases

PROJECTS

In-Silico Docking: Designed ligands for muscarinic acetylcholine receptor M1 using AutoDock

Antimicrobial Study: Tested green algae (*Turbinaria Ornata*) extract for antibacterial activity

NGS & Neuroinformatics: Bash/Python-based sequence analysis on neuroreceptor genes

Bioinformatics Platform: Django-based clinic/attendance/payment management system on AWS

CERTIFICATIONS

Concepts of clinical research, Advanced Diploma in Bioinformatics, Regulatory Affairs, Good Laboratory practices and Good Manufacturing Practices, Lab automation & artificial intelligence in life sciences.

EDUCATION

MSc Biotechnology – Sri Sankara College, University of Madras | 2019 – 2021

Core Modules: Molecular Biology, Genetic Engineering, Immunotechnology, Bioinformatics, Microbial Technology, Cell & Tissue Culture, Bioprocess Engineering, Clinical Research & Regulatory Affairs, Biostatistics.

BSc Biotechnology – Sri Sankara College, University of Madras | 2016 – 2019

Core Modules: Microbiology, Biochemistry, Molecular Genetics, Environmental Biotechnology, Enzyme Technology, Plant and Animal Biotechnology, Instrumentation Techniques.

PRESENTATIONS & ACTIVITIES

Poster: Bioplastics (PHB), PPT: Dairy Factory Blueprint | Organizer: Omics Conference, IV trip to ICAR, Kozhikode